Ali Siahkoohi

Assistant Professor
Department of Computer Science
University of Central Florida

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https://alisiahkoohi.github.io
Last updated: October 14, 2026

Research Interests

My research focuses on uncertainty quantification in computational science and engineering, specifically high-dimensional probabilistic modeling and inference for physical systems.

Academic Appointments

University of Central Florida — August 2025 – Present
Tenure-Track Assistant Professor — Orlando, FL, USA
Department of Computer Science

Rice University — August 2022 – July 2025
Simons Postdoctoral Fellow — Houston, TX, USA
Department of Computational Applied Mathematics & Operations Research
Jointly hosted by Maarten V. de Hoop and Richard G. Baraniuk

Education

Georgia Institute of Technology — August 2022
Ph.D. in Computational Science and Engineering — Atlanta, GA, USA
Advised by Felix J. Herrmann

University of Tehran — March 2016
M.Sc. in Geophysics — Tehran, Iran

Sharif University of Technology — August 2013
B.Sc. in Electrical Engineering — Tehran, Iran

Publications

Google Scholar profile: https://scholar.google.com/citations?user=sxRMqYIAAAAJ&h

In Preparation & Under Review

P6. A. Vishwakarma, K. Aghazade, A. Siahkoohi, and A. Gholami. Distributed proximal Stein variational gradient descent algorithm for large-scale Bayesian inference in traveltime tomography. Preprint arXiv:2609.26653, 2026
[pdf] [code] [bib]

P5. A. Siahkoohi and S. Alemohammad. Priors learned from legacy reconstructions inherit undetectable overconfidence. Preprint arXiv:2607.21721, 2026
[pdf] [code] [bib]

P4. A. Siahkoohi. Amortized quadrature for posterior expectations in inverse problems. Preprint arXiv:2606.15871, 2026a
[pdf] [code] [bib]

P3. A. Siahkoohi. A lift for input-convex neural net training. Preprint arXiv:2605.24274, 2026b
[pdf] [code] [bib]

P2. A. Siahkoohi, K. Aghazade, and A. Gholami. Dual-space posterior sampling for Bayesian inference in constrained inverse problems. Preprint arXiv:2603.00393, 2026a

[pdf] [poster] [code] [bib]

P1. A. Thatipelli and A. Siahkoohi. Hypernetwork-based approach for grid-independent functional data clustering. Preprint arXiv:2602.22823, 2026
[pdf] [poster] [code] [bib]

Journal Publications

J10. A. Thatipelli, J. Sam, M. Louboutin, A. Siahkoohi, R. Wang, and F. J. Herrmann. XConv: Low-memory stochastic backpropagation for convolutional layers. ***Transactions on Machine Learning Research***, 2026
[pdf] [slides] [video] [code] [link] [bib]

J9. A. Siahkoohi, R. Morel, R. Balestriero, E. Allys, G. Sainton, T. Kawamura, and M. V. de Hoop. Multiscale clustering and source separation of InSight mission seismic data. ***IEEE Transactions on Geoscience and Remote Sensing***, 64:1–16, 2026b
[pdf] [slides] [code] [link] [bib]

J8. R. Orozco, A. Siahkoohi, M. Louboutin, and F. J. Herrmann. ASPIRE: Iterative amortized posterior inference for Bayesian inverse problems. ***Inverse Problems***, 41(4):045001, 2025
[pdf] [code] [link] [bib]

J7. R. Orozco, P. Witte, M. Louboutin, A. Siahkoohi, G. Rizzuti, B. Peters, and F. J. Herrmann. InvertibleNetworks.jl: A Julia package for scalable normalizing flows. ***Journal of Open Source Software***, 9(99):6554, 2024
[pdf] [code] [link] [bib]

J6. L. Luzi, P. M. Mayer, J. Casco-Rodriguez, A. Siahkoohi, and R. G. Baraniuk. Boomerang: Local sampling on image manifolds using diffusion models. ***Transactions on Machine Learning Research***, 2024a
[pdf] [code] [link] [bib]

J5. M. Louboutin, Z. Yin, R. Orozco, T. J. Grady II, A. Siahkoohi, G. Rizzuti, P. A. Witte, O. Møyner, G. J. Gorman, and F. J. Herrmann. Learned multiphysics inversion with differentiable programming and machine learning. ***The Leading Edge***, 42(7):474–486, 2023a
[pdf] [link] [bib] [featured in Seismic Soundoff] [journal's most downloaded paper in '23]

J4. Y. Zhang, Z. Yin, O. López, A. Siahkoohi, M. Louboutin, R. Kumar, and F. J. Herrmann. Optimized time-lapse acquisition design via spectral gap ratio minimization. ***Geophysics***, 88(4):A19–A23, 2023a
[pdf] [link] [bib]

J3. A. Siahkoohi, G. Rizzuti, R. Orozco, and F. J. Herrmann. Reliable amortized variational inference with physics-based latent distribution correction. ***Geophysics***, 88(3):R297–R322, 2023a
[pdf] [slides] [code] [link] [bib] [featured in Geophysics Bright Spots]

J2. A. Siahkoohi, G. Rizzuti, and F. J. Herrmann. Deep Bayesian inference for seismic imaging with tasks. ***Geophysics***, 87(5):S281–S302, 2022a
[pdf] [code] [link] [bib]

J1. A. Siahkoohi, M. Louboutin, and F. J. Herrmann. The importance of transfer learning in seismic modeling and imaging. ***Geophysics***, 84(6):A47–A52, 2019a
[pdf] [code] [link] [bib]

Peer-Reviewed Conference Proceedings

C36. A. Siahkoohi and H. Oh. Conditional neural control variates for variance reduction in Bayesian inverse problems. In ***Proceedings of the Forty-second Conference on Uncertainty in Artificial Intelligence***, volume 337, pages 6321–6341, 2026
[pdf] [poster] [code] [link] [bib]

C35. A. Siahkoohi and D. Sabeddu. On the role of memorization in learned priors for geophysical inverse problems. In ***Society of Exploration Geophysicists Technical Program Expanded Abstracts***, 2026. In print

[pdf] [slides] [code] [bib]

C34. K. Aghazade, A. Siahkoohi, and A. Gholami. Scalable Bayesian full waveform inversion via dual augmented Lagrangian SVGD. In ***Society of Exploration Geophysicists Technical Program Expanded Abstracts***, 2026. In print
[pdf] [bib]

C33. S. Alemohammad, J. Casco-Rodriguez, L. Luzi, A. I. Humayun, H. Babaei, D. LeJeune, A. Siahkoohi, and R. G. Baraniuk. Self-consuming generative models go MAD. In *The Twelfth* ***International Conference on Learning Representations***, 2024
[pdf] [extended pdf] [poster] [link] [bib] [featured in the news [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13]]

C32. L. Luzi, D. LeJeune, A. Siahkoohi, S. Alemohammad, V. Saragadam, H. Babaei, N. Liu, Z. Wang, and R. G. Baraniuk. Titan: Bringing the deep image prior to implicit representations. In ***IEEE International Conference on Acoustics, Speech and Signal Processing***, pages 6165–6169, 2024b
[pdf] [code] [link] [bib]

C31. L. Baldassari, A. Siahkoohi, J. Garnier, K. Sølna, and M. V. de Hoop. Conditional score-based diffusion models for Bayesian inference in infinite dimensions. In ***Advances in Neural Information Processing Systems***, volume 36, pages 24262–24290, 2023
[pdf] [slides] [poster] [code] [link] [bib] [featured as a Spotlight presentation]

C30. A. Siahkoohi, R. Morel, M. V. de Hoop, E. Allys, G. Sainton, and T. Kawamura. Unearthing InSights into Mars: Unsupervised source separation with limited data. In ***Proceedings of the 40th International Conference on Machine Learning***, volume 202, pages 31754–31772, 2023b
[pdf] [slides] [poster] [code] [link] [bib]

C29. R. Orozco, M. Louboutin, A. Siahkoohi, G. Rizzuti, T. van Leeuwen, and F. J. Herrmann. Amortized normalizing flows for transcranial ultrasound with uncertainty quantification. In ***Medical Imaging with Deep Learning***, volume 227, pages 332–349, 2023a
[pdf] [link] [bib]

C28. R. Orozco, A. Siahkoohi, M. Louboutin, and F. J. Herrmann. Refining amortized posterior approximations using gradient-based summary statistics. In ***5th Symposium on Advances in Approximate Bayesian Inference***, 2023b
[pdf] [link] [bib]

C27. R. Orozco, A. Siahkoohi, G. Rizzuti, T. van Leeuwen, and F. J. Herrmann. Adjoint operators enable fast and amortized machine learning based Bayesian uncertainty quantification. In ***Medical Imaging 2023: Image Processing***, volume 12464, page 124641L, 2023c
[pdf] [link] [bib]

C26. M. Louboutin, R. Orozco, A. Siahkoohi, and F. J. Herrmann. Learned one-shot imaging. In ***Third International Meeting for Applied Geoscience & Energy***, 2023b
[html] [poster] [code] [bib]

C25. Y. Zhang, Z. Yin, O. Lopez, A. Siahkoohi, M. Louboutin, and F. J. Herrmann. 3D seismic survey design by maximizing the spectral gap. In ***Third International Meeting for Applied Geoscience & Energy***, 2023b
[pdf] [poster] [bib]

C24. A. Siahkoohi, M. Chinen, T. Denton, W. B. Kleijn, and J. Skoglund. Ultra-low-bitrate speech coding with pretrained Transformers. In ***Proceedings of Interspeech***, pages 4421–4425, 2022b
[pdf] [link] [bib]

C23. A. Siahkoohi, M. Louboutin, and F. J. Herrmann. Velocity continuation with Fourier neural operators for accelerated uncertainty quantification. In ***Society of Exploration Geophysicists Technical Program Expanded Abstracts***, pages 1765–1769, 2022c
[pdf] [slides] [code] [link] [bib]

C22. M. Louboutin, P. Witte, A. Siahkoohi, G. Rizzuti, Z. Yin, R. Orozco, and F. J. Herrmann. Accelerating

innovation with software abstractions for scalable computational geophysics. In ***Society of Exploration Geophysicists Technical Program Expanded Abstracts***, pages 1482–1486, 2022
[pdf] [slides] [link] [bib]

C21. Z. Yin, A. Siahkoohi, M. Louboutin, and F. J. Herrmann. Learned coupled inversion for carbon sequestration monitoring and forecasting with Fourier neural operators. In ***Society of Exploration Geophysicists Technical Program Expanded Abstracts***, pages 467–472, 2022
[pdf] [slides] [code] [link] [bib] [student oral paper honorable mention]

C20. Y. Zhang, M. Louboutin, A. Siahkoohi, Z. Yin, R. Kumar, and F. J. Herrmann. A simulation-free seismic survey design by maximizing the spectral gap. In ***Society of Exploration Geophysicists Technical Program Expanded Abstracts***, pages 15–20, 2022
[pdf] [slides] [code] [link] [bib]

C19. A. Siahkoohi, R. Orozco, G. Rizzuti, and F. J. Herrmann. Wave-equation based inversion with amortized variational Bayesian inference. In ***EAGE Deep learning for seismic processing: Investigating the foundations workshop***, 2022d
[pdf] [slides] [code] [link] [bib]

C18. R. Orozco, A. Siahkoohi, G. Rizzuti, T. van Leeuwen, and F. J. Herrmann. Photoacoustic imaging with conditional priors from normalizing flows. In ***NeurIPS Workshop on Deep Learning and Inverse Problems***, 2021
[pdf] [poster] [link] [bib]

C17. A. Siahkoohi, G. Rizzuti, M. Louboutin, P. Witte, and F. J. Herrmann. Preconditioned training of normalizing flows for variational inference in inverse problems. In ***3rd Symposium on Advances in Approximate Bayesian Inference***, 2021
[pdf] [slides] [code] [link] [bib]

C16. A. Siahkoohi and F. J. Herrmann. Learning by example: Fast reliability-aware seismic imaging with normalizing flows. In ***Society of Exploration Geophysicists Technical Program Expanded Abstracts***, pages 1580–1585, 2021
[pdf] [slides] [code] [link] [bib]

C15. R. Kumar, M. Kotsi, A. Siahkoohi, and A. Malcolm. Enabling uncertainty quantification for seismic data preprocessing using normalizing flows (NF)—An interpolation example. In ***Society of Exploration Geophysicists Technical Program Expanded Abstracts***, pages 1515–1519, 2021
[pdf] [code] [link] [bib]

C14. G. Rizzuti, A. Siahkoohi, P. A. Witte, and F. J. Herrmann. Parameterizing uncertainty by deep invertible networks, an application to reservoir characterization. In ***Society of Exploration Geophysicists Technical Program Expanded Abstracts***, pages 1541–1545, 2020
[pdf] [slides] [code] [link] [bib]

C13. M. Zhang, A. Siahkoohi, and F. J. Herrmann. Transfer learning in large-scale ocean bottom seismic wavefield reconstruction. In ***Society of Exploration Geophysicists Technical Program Expanded Abstracts***, pages 1666–1670, 2020
[pdf] [slides] [code] [link] [bib]

C12. A. Siahkoohi, G. Rizzuti, and F. J. Herrmann. Weak deep priors for seismic imaging. In ***Society of Exploration Geophysicists Technical Program Expanded Abstracts***, pages 2998–3002, 2020a
[pdf] [slides] [code] [link] [bib]

C11. A. Siahkoohi, G. Rizzuti, and F. J. Herrmann. Uncertainty quantification in imaging and automatic horizon tracking—A Bayesian deep-prior based approach. In ***Society of Exploration Geophysicists Technical Program Expanded Abstracts***, pages 1636–1640, 2020b
[pdf] [slides] [code] [link] [bib]

C10. A. Siahkoohi, G. Rizzuti, and F. J. Herrmann. A deep-learning based Bayesian approach to seismic imaging and uncertainty quantification. In ***European Association of Geoscientists & Engineers Conference and Exhibition Extended Abstracts***, 2020c

[pdf] [slides] [code] [link] [bib]

C9. F. J. Herrmann, A. Siahkoohi, and G. Rizzuti. Learned imaging with constraints and uncertainty quantification. In ***NeurIPS Deep Inverse Workshop***, 2019
[pdf] [slides] [poster] [link] [bib]

C8. A. Siahkoohi, R. Kumar, and F. J. Herrmann. Deep-learning based ocean bottom seismic wavefield recovery. In ***Society of Exploration Geophysicists Technical Program Expanded Abstracts***, pages 2232–2237, 2019b
[pdf] [slides] [code] [link] [bib]

C7. A. Siahkoohi, D. J. Verschuur, and F. J. Herrmann. Surface-related multiple elimination with deep learning. In ***Society of Exploration Geophysicists Technical Program Expanded Abstracts***, pages 4629–4634, 2019c
[pdf] [slides] [link] [bib]

C6. G. Rizzuti, A. Siahkoohi, and F. J. Herrmann. Learned iterative solvers for the Helmholtz equation. In ***European Association of Geoscientists & Engineers Conference and Exhibition Extended Abstracts***, 2019
[pdf] [slides] [link] [bib]

C5. A. Siahkoohi, M. Louboutin, R. Kumar, and F. J. Herrmann. Deep convolutional neural networks in prestack seismic—two exploratory examples. In ***Society of Exploration Geophysicists Technical Program Expanded Abstracts***, pages 2196–2200, 2018a
[pdf] [poster] [link] [bib]

C4. A. Siahkoohi, R. Kumar, and F. J. Herrmann. Seismic data reconstruction with generative adversarial networks. In ***European Association of Geoscientists & Engineers Conference and Exhibition Extended Abstracts***, 2018b
[pdf] [slides] [link] [bib]

C3. A. Siahkoohi and A. Gholami. Sparsity promoting least squares migration for laterally inhomogeneous media. In ***7th EAGE Saint Petersburg International Conference and Exhibition***, 2016
[pdf] [link] [bib]

C2. M. S. Ebrahimi, M. H. Daraei, J. Rezaei, and A. Siahkoohi. A novel utilization of wireless sensor networks as data acquisition system in smart grids. In ***Materials Science and Information Technology***, volume 433-440, pages 6725–6730, 2012
[pdf] [link] [bib]

C1. A. Najafi, A. Siahkoohi, and M. B. Shamsollahi. A content-based digital image watermarking algorithm robust against JPEG compression. In ***IEEE International Symposium on Signal Processing and Information Technology***, pages 432–437, 2011
[pdf] [link] [bib]

Theses

T1. A. Siahkoohi. *Deep generative models for solving geophysical inverse problems*. PhD thesis, **Georgia Institute of Technology**, 2022
[pdf] [slides] [link] [bib]

Technical Reports

R4. P. M. Mayer, L. Luzi, A. Siahkoohi, D. H. Johnson, and R. G. Baraniuk. Improving fairness and mitigating MADness in generative models. Technical Report arXiv:2405.13977, Rice University, 2024
[pdf] [slides] [code] [bib]

R3. L. Baldassari, A. Siahkoohi, J. Garnier, K. Sølna, and M. V. de Hoop. Taming score-based diffusion priors for infinite-dimensional nonlinear inverse problems. Technical Report arXiv:2405.15676, Rice University, 2024
[pdf] [bib]

R2. A. Siahkoohi, G. Rizzuti, P. A. Witte, and F. J. Herrmann. Faster uncertainty quantification for inverse problems with conditional normalizing flows. Technical Report arXiv:2007.07985, Georgia Institute of Technology, 2020d [pdf] [link] [bib]

R1. A. Siahkoohi, M. Louboutin, and F. J. Herrmann. Neural network augmented wave-equation simulation. Technical Report arXiv:1910.00925, Georgia Institute of Technology, 2019d
[pdf] [code] [link] [bib]

Awards

Future Faculty Fellows Award June 2024
Rice University, George R. Brown School of Engineering and Computing Houston, TX, USA
[link]

Mentoring Experience

University of Central Florida Orlando, FL, USA
Departments of Computer Science & Electrical and Computer Engineering

- Banafsheh Adami [link] 2026 – Present
  CS Ph.D. Student — co-advised with Alain Kassab
- Paul Hawkins [link] 2026 – Present
  CS Undergraduate Student — Undergraduate thesis advisor
- Anirudh Thatipelli [link] 2025 – Present
  CS Ph.D. Student — Ph.D. advisor
- Davide Sabeddu [link] 2025 – 2026
  ECE Ph.D. Student — Advised on the development of methods for mitigating overfitting in generative models and co-authored a paper (A. Siahkoohi and Sabeddu, 2026)

Rice University Houston, TX, USA
Department of Computational Applied Mathematics & Operations Research

- Jeffrey J. Sam [link] 2024 – 2025
  M.Sc. Student — Advised on the design and implementation of experiments and co-authored a paper (Thatipelli et al., 2026)
- Paul M. Mayer [link] 2022 – 2025
  Ph.D. Student — Advised on the development of methods and software for two projects and co-authored two papers (Luzi et al., 2024a; Mayer et al., 2024)

Georgia Institute of Technology Atlanta, GA, USA
School of Computational Science and Engineering

- Rafael Orozco [link] 2020 – 2022
  Ph.D. Student — Advised on the development of methods and software for main Ph.D. thesis and co-authored four papers (Orozco et al., 2021, 2023b,c, 2025)
- Mi Zhang [link] 2019 – 2020
  Visiting Ph.D. Student (China University of Petroleum-Beijing) — Advised on the development of methods and software for a project and co-authored a paper (Zhang et al., 2020)

Teaching Experience

University of Central Florida Orlando, FL, USA
Department of Computer Science

- Sampling Methods for Uncertainty Quantification [link] Spring 2027
  Instructor of Record

- Algorithms for Machine Learning [link] — Fall 2026
  Instructor of Record
- Sampling Methods for Uncertainty Quantification [link] — Spring 2026
  Instructor of Record
- Algorithms for Machine Learning [link] — Fall 2025
  Instructor of Record

Rice University — Houston, TX, USA
Department of Computational Applied Mathematics & Operations Research

- Numerical Analysis — Fall 2024
  Substitute Instructor (12 lectures)
- Numerical Analysis I — Fall 2022
  Substitute Instructor (18 lectures)

Georgia Institute of Technology — Atlanta, GA, USA
School of Computational Science and Engineering

- Computational Foundations of Machine Learning — Spring 2022
  Teaching Assistant
- Imaging with Data-Driven Models — Fall 2019
  Teaching Assistant
- Numerical Analysis I — Fall 2018
  Teaching Assistant

Sharif University of Technology — Tehran, Iran
Department of Electrical Engineering

- Digital Signal Processing — Spring 2011
  Teaching Assistant
- Signals and Systems — Spring 2011
  Teaching Assistant
- Linear Algebra — Spring 2010
  Teaching Assistant
- Electrical Engineering: Principles and Laboratory — Fall 2009
  Teaching Assistant

Talks

Invited Talks

T35. **University of Central Florida** — April 2026
On the role of memorization in learned priors for geophysical inverse problems — Oral presentation
Foundations of AI Seminar

T34. **IEEE Computer Society, Chapter of San Diego** — June 2025
Mitigating biases in self-consuming generative models — Virtual oral presentation
Open Research Institute (ORI)
[video]

T33. **University of Central Florida** — April 2025
Towards reliable AI: A framework for quantification of AI uncertainty — Oral presentation
Department of Computer Science

T32. **Montana State University** — March 2025
Towards reliable AI: A framework for quantification of AI uncertainty — Oral presentation
Gianforte School of Computing

T31. **The University of California, Santa Barbara** — February 2025
Towards reliable AI: A framework for quantification of AI uncertainty — Oral presentation
Department of Mechanical Engineering

T30. **Johns Hopkins University** — January 2025
Towards reliable AI: A framework for quantification of AI uncertainty — Oral presentation
Department of Electrical and Computer Engineering

T29. **ISCL Seminar Series, Pennsylvania State University** — November 2024
Mitigating biases in self-consuming generative models — Virtual oral presentation
Interdisciplinary Scientific Computing Laboratory (Dr. Romit Maulik)
[video]

T28. **CNRS, Université Montpellier** — January 2023
Low-cost uncertainty quantification for large-scale inverse problems — Virtual oral presentation
RhEoVOLUTION Group (Dr. Andréa Tommasi)

T27. **Workshop on Subsurface Uncertainty Description and Estimation** — August 2022
Reliable amortized variational inference with conditional normalizing flows via physics-based latent distribution correction — Oral presentation
International Meeting for Applied Geoscience & Energy

T26. **Intelligent illumination of the Earth Workshop** — June 2021
Fast and reliability-aware seismic imaging with conditional normalizing flows — Virtual oral presentation
King Abdullah University of Science and Technology

T25. **Advances in Seismic Imaging and Inversion Mini-symposium** — October 2020
Unsupervised data-guided uncertainty analysis in imaging and horizon tracking — Virtual oral presentation
The 3rd Annual Meeting of the SIAM Texas–Louisiana Section

Contributed Talks

T24. **International Meeting for Applied Geoscience & Energy** — August 2026
On the role of memorization in learned priors for geophysical inverse problems — Oral presentation

T23. **International Meeting for Applied Geoscience & Energy** — August 2026
Scalable Bayesian full waveform inversion via dual augmented Lagrangian SVGD — Oral presentation

T22. **Conference on Uncertainty in Artificial Intelligence** — August 2026
Conditional neural control variates for variance reduction in Bayesian inverse problems — Poster presentation

T21. **SIAM Annual Meeting** — July 2026
Conditional neural control variates for variance reduction in Bayesian inverse problems — Poster presentation

T20. **UCF Spring AI Research Day, UCF Institute of Artificial Intelligence** — February 2026
Multi-scale clustering and source separation of InSight mission seismic data — Oral presentation

T19. **Geo-Mathematical Imaging Group Partners Meeting, Rice University** — November 2024
Improving fairness and mitigating MADness in generative models — Oral presentation

T18. **International Conference on Machine Learning** — July 2023
Unearthing InSights into Mars: Unsupervised source separation with limited data — Poster presentation

T17. **Symposium on Advances in Approximate Bayesian Inference** — July 2023
Refining amortized posterior approximations using gradient-based summary statistics — Poster presentation

T16. **Geo-Mathematical Imaging Group Partners Meeting, Rice University** — May 2023
Martian time-series unraveled: A multi-scale nested approach with factorial variational autoencoders — Oral presentation

T15. **Geo-Mathematical Imaging Group Partners Meeting, Rice University** May 2023
Unearthing InSights into Mars: Unsupervised source separation with limited data Oral presentation

T14. **International Meeting for Applied Geoscience & Energy** August 2022
Velocity continuation with Fourier neural operators for accelerated uncertainty quantification Oral presentation

T13. **Chrome Media Team, Google** December 2021
Low-bitrate speech coding with Transformers Virtual oral presentation

T12. **ML4SEISMIC Partners Meeting, Georgia Institute of Technology** November 2021
Multifidelity conditional normalizing flows for physics-guided Bayesian inference Virtual oral presentation

T11. **ML4SEISMIC Partners Meeting, Georgia Institute of Technology** November 2021
Uncertainty quantification in imaging and automatic horizon tracking—A Bayesian deep-prior based approach Virtual oral presentation

T10. **Society of Exploration Geophysicists International Exposition and Annual Meeting** September 2021
Learning by example: Fast reliability-aware seismic imaging with normalizing flows Virtual oral presentation
[video]

T9. **Symposium on Advances in Approximate Bayesian Inference** January 2021
Preconditioned training of normalizing flows for variational inference in inverse problems Prerecorded short oral presentation
[video]

T8. **European Association of Geoscientists & Engineers Annual Conference & Exhibition** December 2020
A deep-learning based Bayesian approach to seismic imaging and uncertainty quantification Virtual oral presentation

T7. **Society of Exploration Geophysicists International Exposition and Annual Meeting** October 2020
Uncertainty quantification in imaging and automatic horizon tracking—A Bayesian deep-prior based approach Virtual oral presentation
[video]

T6. **Society of Exploration Geophysicists International Exposition and Annual Meeting** October 2020
Weak deep priors for seismic imaging Virtual oral presentation
[video]

T5. **Society of Exploration Geophysicists Student Chapter, Georgia Tech** February 2020
A deep-learning based Bayesian approach to seismic imaging and uncertainty quantification Oral presentation

T4. **HotCSE Seminar, CSE Department, Georgia Institute of Technology** November 2019
Learned imaging with constraints and uncertainty quantification Oral presentation

T3. **Society of Exploration Geophysicists International Exposition & Annual Meeting** September 2019
Deep-learning based ocean bottom seismic wavefield recovery Oral presentation

T2. **Society of Exploration Geophysicists International Exposition & Annual Meeting** September 2019
Surface-related multiple elimination with deep learning Oral presentation

T1. **Society of Exploration Geophysicists International Exposition & Annual Meeting** October 2018
Deep convolutional neural networks in prestack seismic—two exploratory examples Poster presentation

Professional Service

Editorial Service

- **Acta Geophysica**, Associate Editor — 2024 – Present
  Applied Geophysics section
- **Journal of Mathematics**, Guest Editor — 2023 – 2024
  Special issue on Applied Mathematics in Inverse Problems and Uncertainty Quantification

Conference Service

- **Artificial Intelligence and Statistics Conference (AISTATS)**, Area Chair — 2027
- **International Meeting for Applied Geoscience & Energy**, Session Chair — 2022

Technical Program Committee Member and Reviewer

- International Conference on Learning Representations (ICLR) — 2024 – 2027
- The Conference on Uncertainty in Artificial Intelligence (UAI) — 2026
- Symposium on Probabilistic Machine Learning (ProbML) — 2026
- International Conference on Machine Learning (ICML) — 2024 – 2026
- Annual AAAI Conference on Artificial Intelligence — 2025 – 2026
- Structured Probabilistic Inference & Generative Modeling — 2023 – 2025
- Frontiers in Probabilistic Inference: Sampling Meets Learning — 2025
- Neural Information Processing Systems (NeurIPS) — 2023 – 2025
- Artificial Intelligence and Statistics Conference (AISTATS) — 2024 – 2025
- Advances in Approximate Bayesian Inference (AABI) — 2023 – 2024
- Structured Probabilistic Inference & Generative Modeling (ICML workshop) — 2023 – 2024
- International Speech Communication Association (Interspeech) — 2023
- Deep Generative Models for Health (NeurIPS workshop) — 2023
- International Meeting for Applied Geoscience & Energy — 2023

Journal Reviewer

- Transactions on Machine Learning Research
- IEEE Transactions on Neural Networks and Learning Systems
- IEEE Transactions on Computational Imaging
- Journal of Medical Imaging
- SIAM Journal on Scientific Computing
- Notices of the American Mathematical Society (AMS)
- Entropy
- IEEE Transactions on Geoscience and Remote Sensing
- IEEE Geoscience and Remote Sensing Letters
- Remote Sensing
- Geosciences
- Geophysics
- The Leading Edge
- Geophysical Prospecting
- Journal of Geophysical Research – Solid Earth

Industry Research Experience

Google — August 2021 – December 2021
Research Intern (cf. A. Siahkoohi et al. (2022b)) — San Francisco, CA, USA
Chrome Media Team

Selected Media Coverage

Future Faculty Fellow Ali Siahkoohi joins University of Central Florida as assistant professor June 2025
Rice University Engineering News
[link]

AI's Mad Loops February 2025
Rice Magazine
[link]

AI Appears to Be Slowly Killing Itself August 2024
Futurism
[link]

When A.I.'s Output Is a Threat to A.I. Itself August 2024
The New York Times
[link]

Breaking MAD: Generative AI could break the internet July 2024
Rice News, Rice University
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